

SECR2033
Computer Organization and Architecture

Lecture slides prepared by “Computer Organization and Architecture”, 9/e, by William Stallings, 2013.

Module 7
I/O and Storage Systems

Objectives:

- To understand how **I/O systems** work, including **I/O methods** and **architectures**.
- To familiar with **storage media**, and the **differences** in their respective **formats**.
- To understand how **RAID** improves disk performance and reliability, and which RAID systems are most useful today.

Module 7

I/O and Storage Systems

7.1 Introduction
7.2 I/O Architectures
7.3 Data Storage
7.4 Summary

7.1 Introduction

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- All computers have I/O devices connected to them, and to achieve good performance, I/O should be kept to a minimum!
- In studying I/O, we seek to understand the different types of I/O devices as well as how they work.
- Data storage and retrieval is one of the primary functions of computer systems.
  - One could easily make the argument that computers are more useful to us as data storage and retrieval devices than they are as computational machines.

I/O and Performance

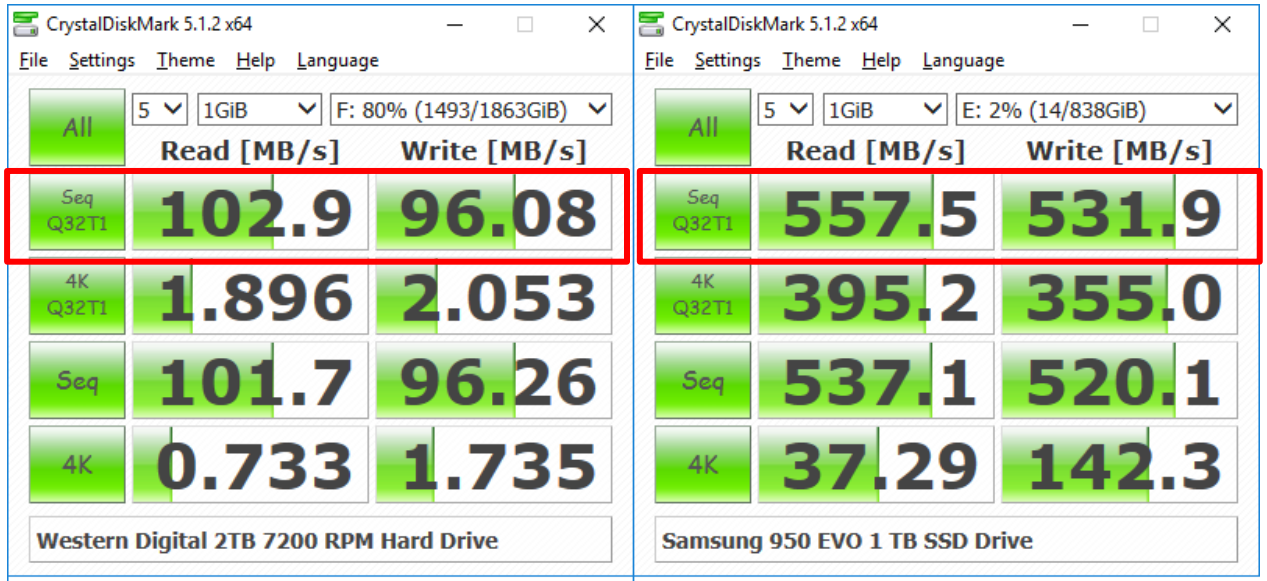
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- I/O performance is measured based on 2 characteristics:
- Data rate (bandwidth) – throughput
  - Amount of data transferred in unit time (units = bytes/time)
  - Critical for application that transfer large amount of data in big blocks (e.g. Bandwidth limiting factor like download large files over crowded WiFi area).
- Latency – I/O response time
  - The time taken for the smallest transfer (unit = time)
  - Primary concern for application that do many small dependent transfers.
- The fastest processor in the world is of little use if it spends most of its time waiting for data.

Bandwidth (bit rate) of network transmission medium

System	Bit Rate	Observations
Telephone twisted pair	33.6-56 kbps	4 kHz telephone channel
Ethernet twisted pair	10 Mbps, 100 Mbps	100 meters of unshielded twisted copper wire pair
Cable modem	500 kbps-4 Mbps	Shared CATV return channel
ADSL twisted pair	64-640 kbps in, 1.536-6.144 Mbps out	Coexists with analog telephone signal
2.4 GHz radio	2-11 Mbps	IEEE 802.11 wireless LAN
28 GHz radio	1.5-45 Mbps	5 km multipoint radio
Optical fiber	2.5-10 Gbps	1 wavelength
Optical fiber	>1600 Gbps	Many wavelengths

Bandwidth of internal HDD: HDD vs SSD vs Flash Drive



Performance Summary:

Hard Drive: 103 MB/sec read, 96 MB/sec write speed
SSD Drive: 558 MB/sec read, 532 MB/sec write speed
M.2 Drive: 2591 MB/sec read, 1544 MB/sec write speed

In Read Performance:

SSD is 5x Faster than HDD

M.2 is 5x Faster than SSD

M.2 is 25x Faster than HDD

Flash Drive:
Samsung EVO (M.2 PCI-e) (*as of 2020*)



Price Tag:

RM 600++ (for 250 GB)

RM 1200++ (for 500 GB)

RM 3200++ (for 1000 GB @ 1TB)



HDD & SSD:
WD 2TB 7200 &
Samsung 850 EVO 1TB (SATA SSD)



Price Tag:

Samsung EVO 850 **1TB** : RM 1250++

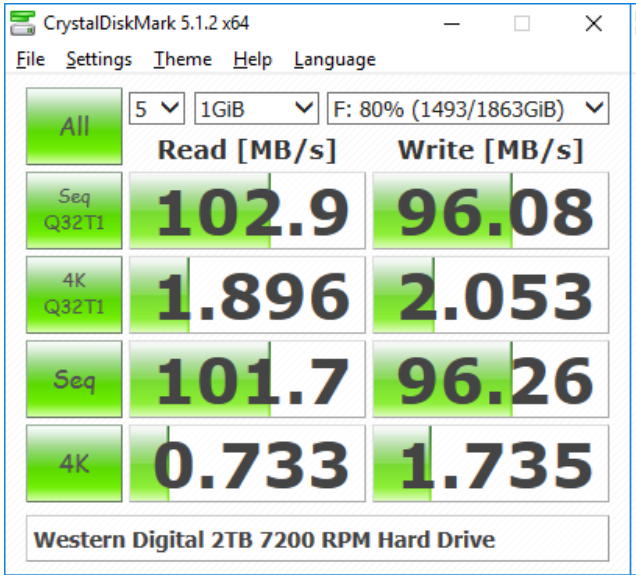
WD 7200 **2TB** : RM 300++



Bandwidth of internal HDD: Price (cost) vs Performance

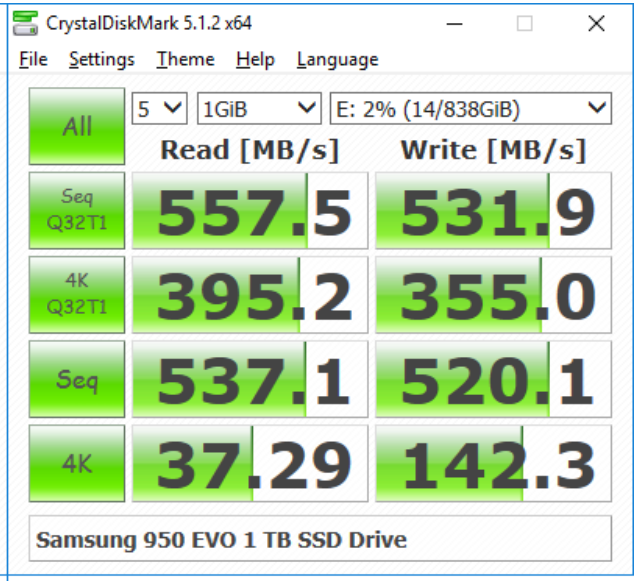
RM 300

2 TB



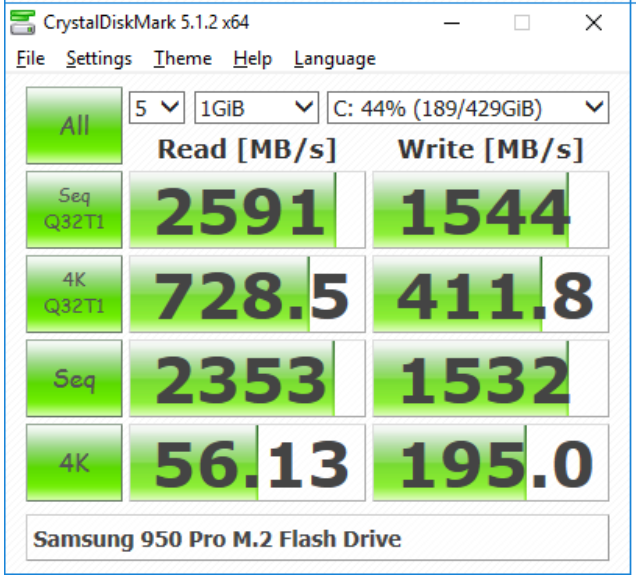
RM 1250

1 TB



RM 3200

1 TB



Performance Summary:

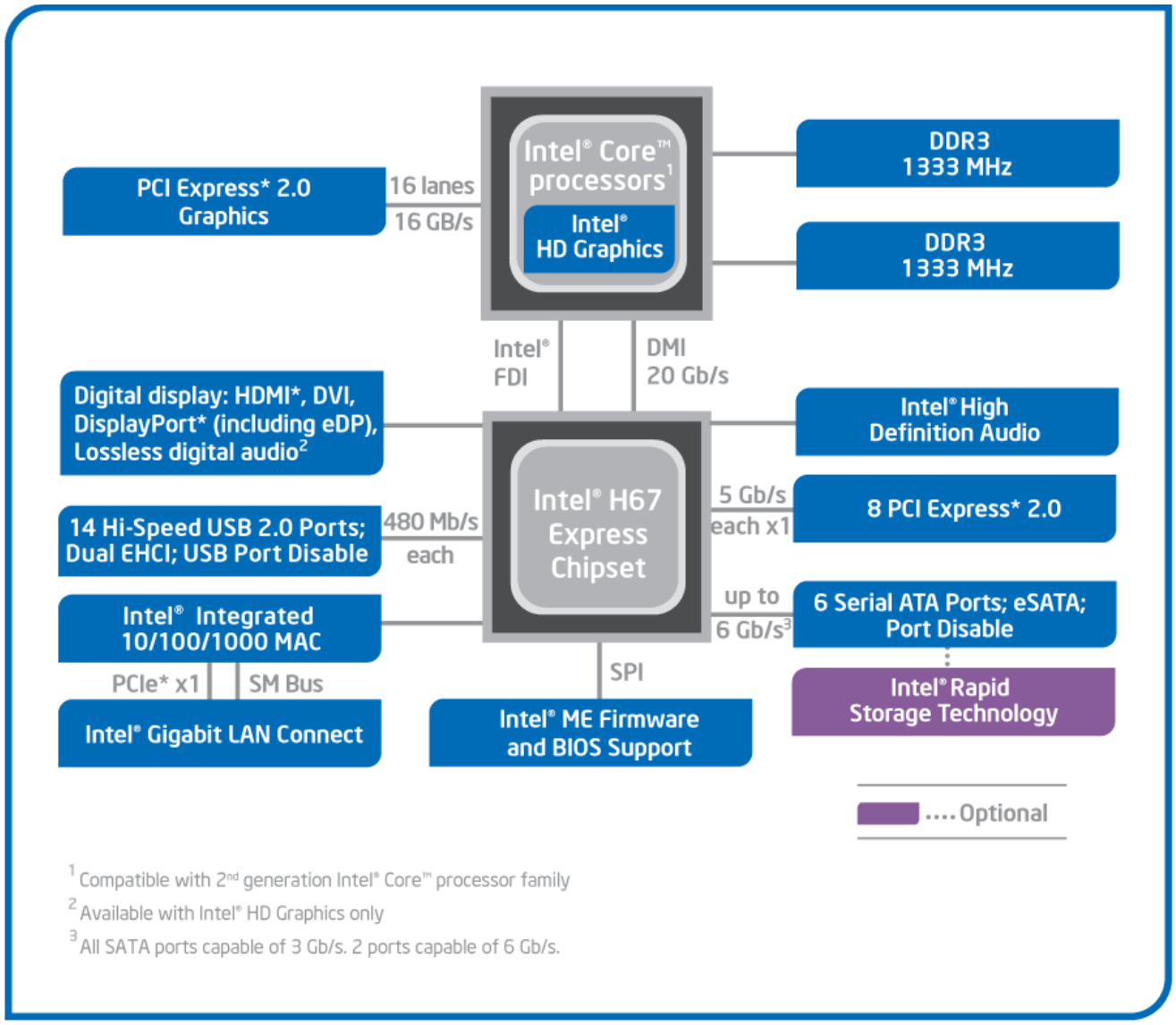
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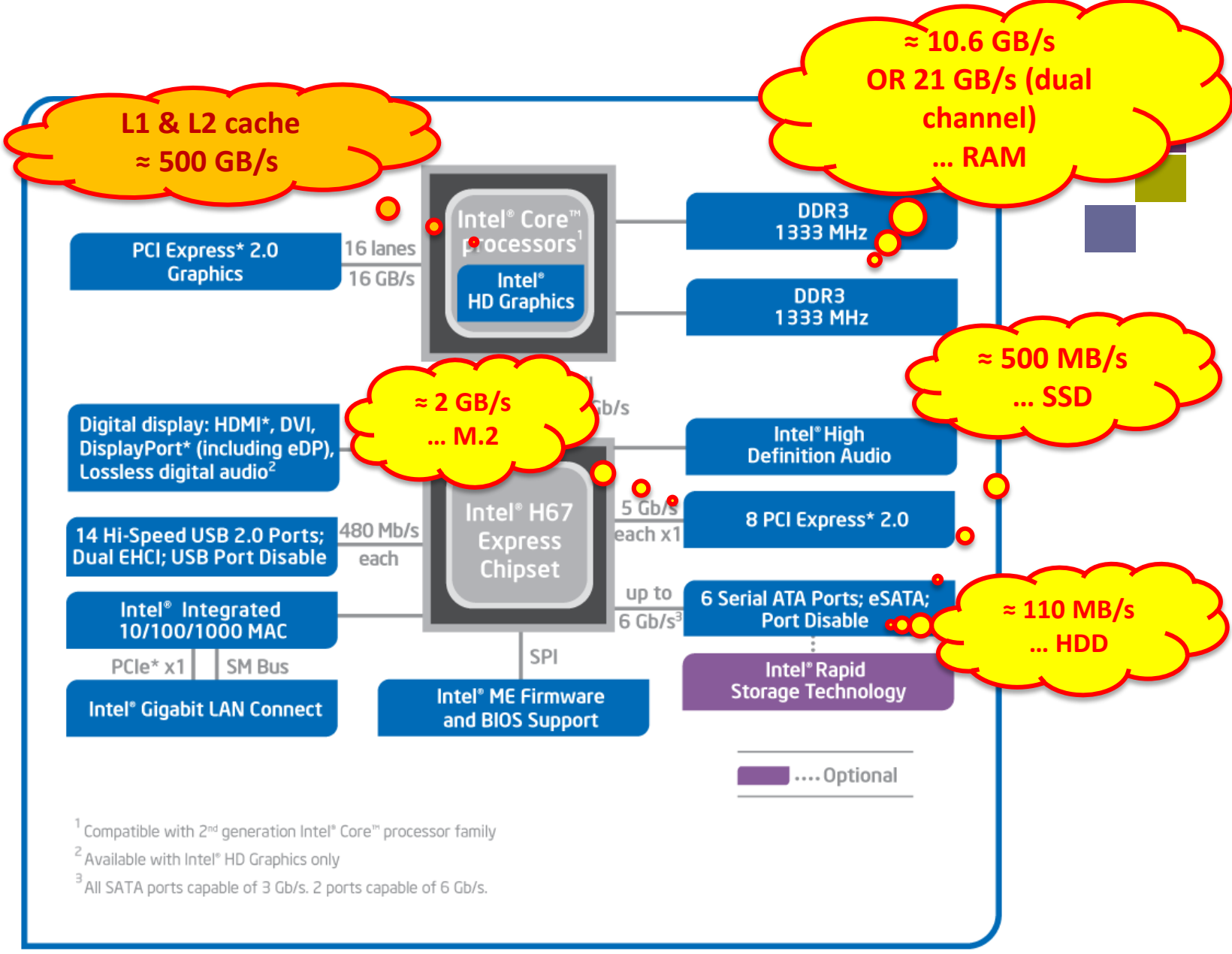
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M.2 is 5x Faster than SSD

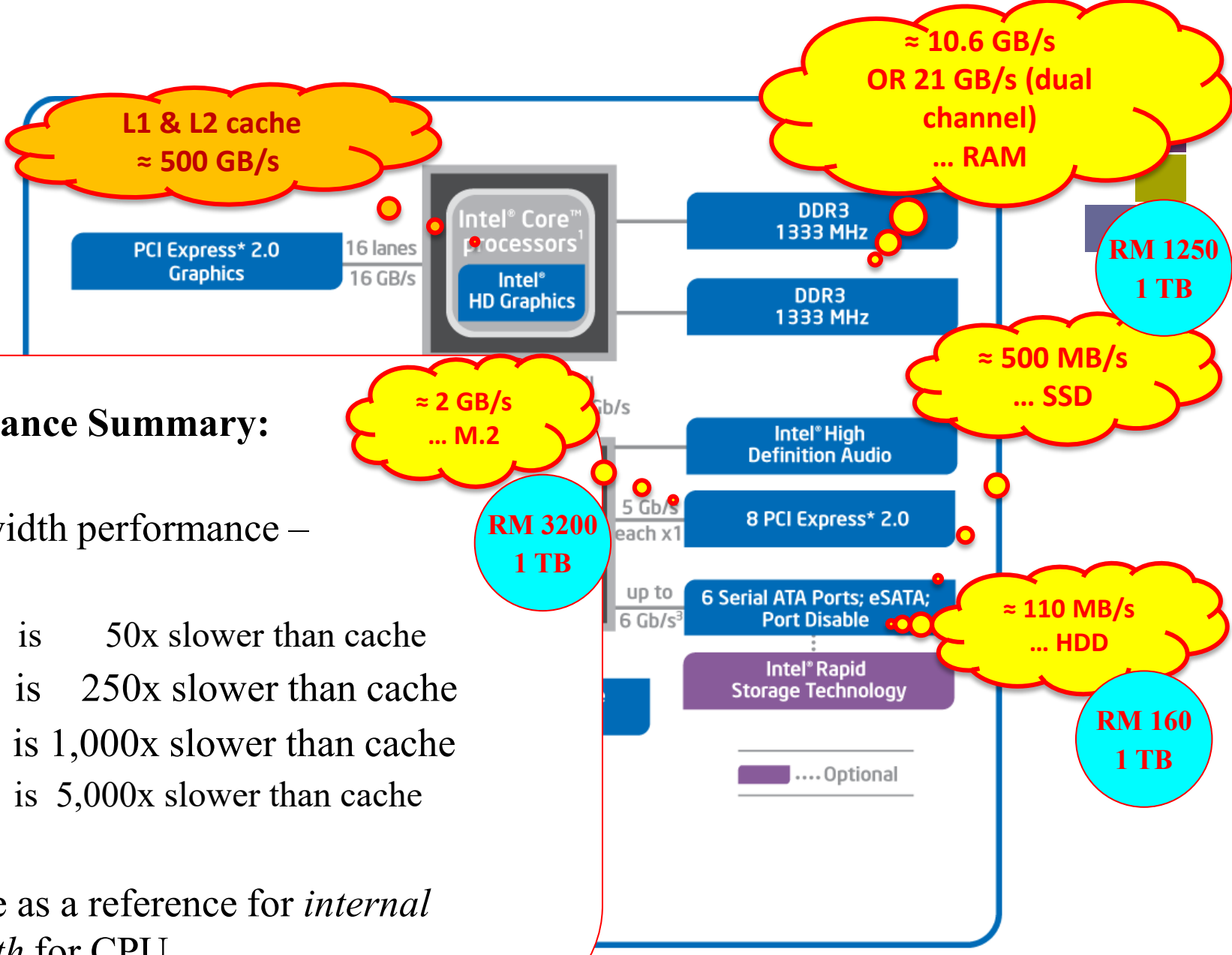
M.2 is 25x Faster than HDD



Intel® H67 Express Chipset Platform Block Diagram



Intel® H67 Express Chipset Platform Block Diagram



Performance Summary:

In bandwidth performance –

- RAM is 50x slower than cache
- M.2 is 250x slower than cache
- SSD is 1,000x slower than cache
- HDD is 5,000x slower than cache

nb: cache as a reference for *internal bandwidth* for CPU

Module 7
I/O and Storage Systems

7.1 Introduction

7.2 I/O Architectures

7.3 Data Storage

7.4 Summary

- Overview
- I/O and Performance
- I/O Control Methods
- I/O Bus Operation

7 Overview

- Input/output (I/O) is defined as a subsystem of components that moves data between external devices and a host system; consisting of a CPU and main memory.
- I/O subsystems include, but are not limited to:

  - Blocks of *main memory* → dedicated to I/O functions.
  - *Buses* → move data into and out of the system.
  - *Control modules* → in the host and in peripheral devices.
  - *Interfaces* to external components → such as keyboards.
  - *Cabling* or communications links → between the host system and its peripherals.

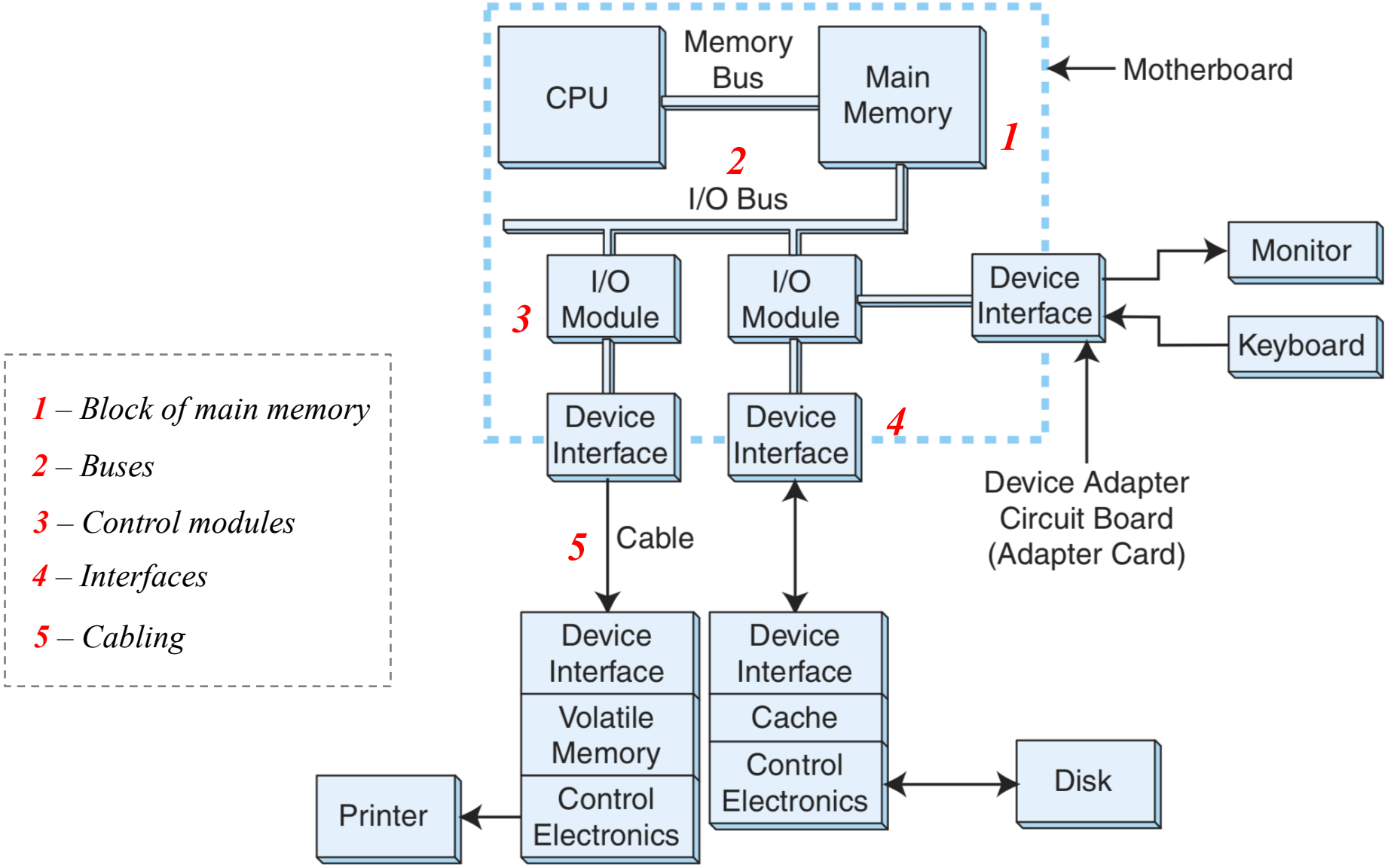


Figure: A model I/O configuration.

7 I/O Control Methods

- Computer systems employ any of four general I/O control methods.

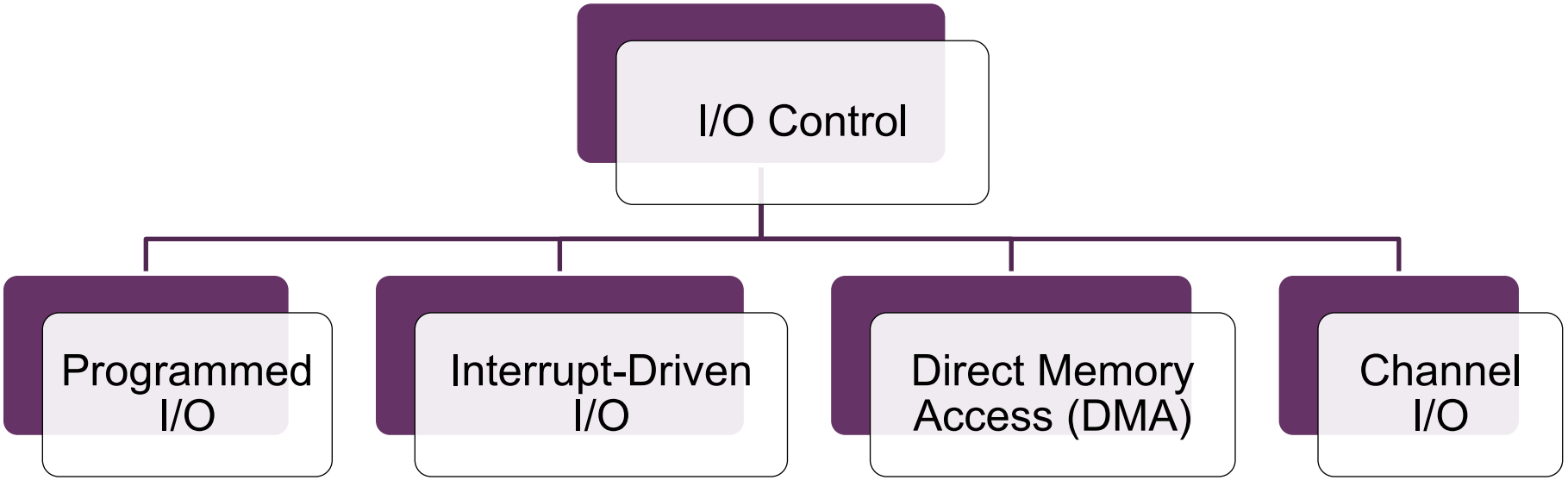
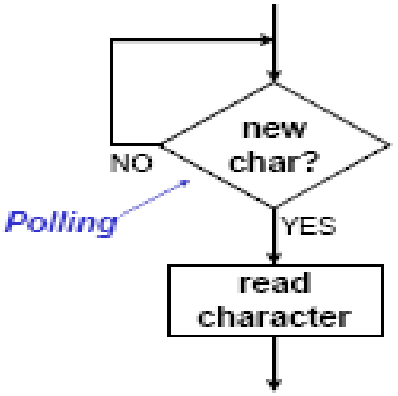


Figure: General I/O control methods

(a) Programmed I/O

- Systems assign at least one register for the exclusive use of each I/O device.
- Programmed I/O sometimes referred *polled I/O* → The CPU continually monitors each register, waiting for data to arrive.
- Example: "Is there new character?"
  - Processor queries I/O device status register
  - Loops until it gets status it wants (ready for next command)
  - Or tries again later



(a) Programmed I/O

- Example usage, best suited for special-purpose systems
  - early ATM (automated teller machines)



- Simple.
- Less hardware.



- Waste of processor's time (processor faster than I/O) – overhead in polling instead of actual data transfer.
- Worse for higher bandwidth I/O devices (*i.e.* disks)

(b) Interrupt-Driven I/O

- Instead of the CPU continually asking its attached devices whether they have any input, the I/O devices tell the CPU when they have data to send (interrupt mechanism).
- Interrupts are usually signalled with a bit in the CPU flags register → *interrupt flag*.

- Once the *interrupt flag* is set, the operating system interrupts (or pause) a program is currently running, saving the program's state and variable information.
- After the CPU has completed servicing the I/O, it restores the information it saved from the program that was running when the interrupt occurred, and the program execution resumes.

INT (Interrupt)
INTA (Interrupt Acknowledge)

1 – Every peripheral device in the system has access to an interrupt request line.

2 – The controller signals the CPU when any of the interrupt lines are asserted.

3 – CPU is ready, assert `INTA` signal.

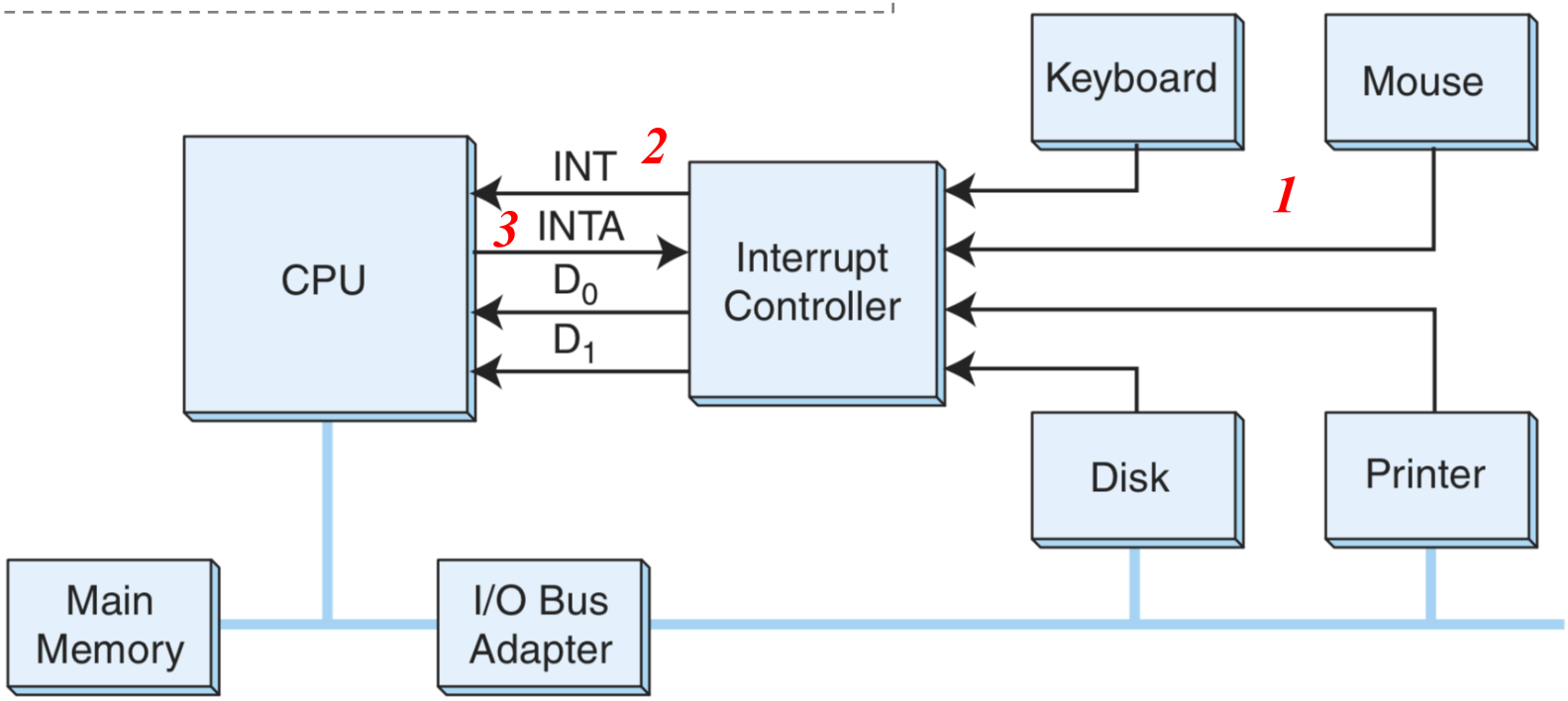


Figure: An I/O subsystem using interrupt.

(c) Direct Memory Access (DMA)

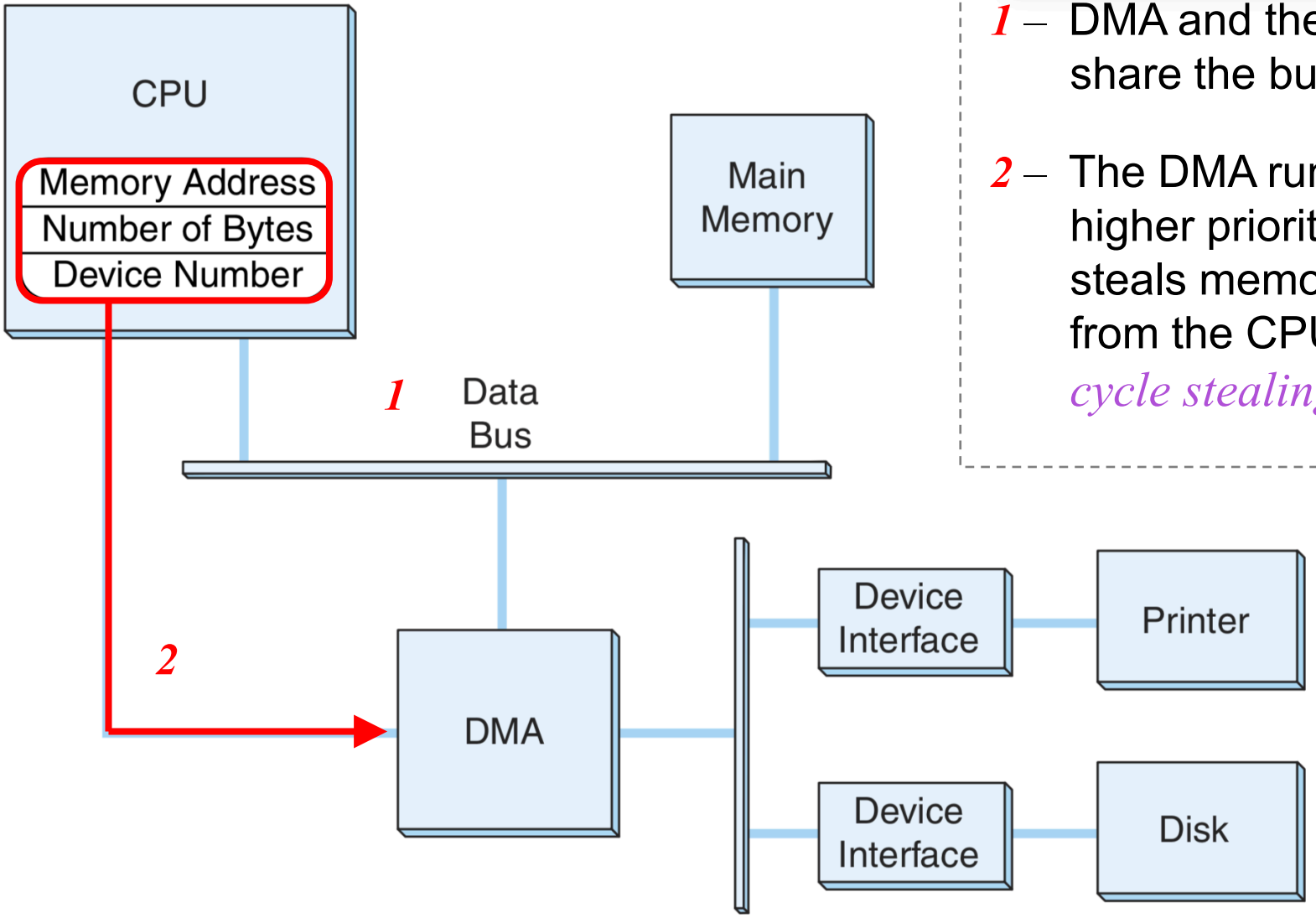
- With both programmed I/O and interrupt-driven I/O, the CPU moves data to / from the I/O device.
- When a system uses DMA, the CPU offloads execution of tedious I/O instructions to DMA controller.

- To effect the transfer, the CPU provides the DMA controller with:

  ✓ the location of the bytes to be transferred,

  ✓ the number of bytes to be transferred, and

  ✓ the destination device or memory address.



1 – DMA and the CPU share the bus.

2 – The DMA runs at a higher priority and steals memory cycles from the CPU (aka *cycle stealing*).

Figure: An example of DMA configuration.

- DMA controller has direct access to the memory while CPU is temporarily disabled
- In DMA, both CPU and DMA controller have direct access to main memory via shared system bus (i.e. data, address and control lines)
- A DMA controller *temporarily borrows* the address bus, data bus and control bus from CPU, and transfers the data bytes directly between an I/O port and a series of memory locations

(d) Channel I/O

- An intelligent type of DMA interface known as an channel I/O.
- DMA I/O requires only a little less CPU participation than does interrupt-driven I/O.
- DMA has overhead:

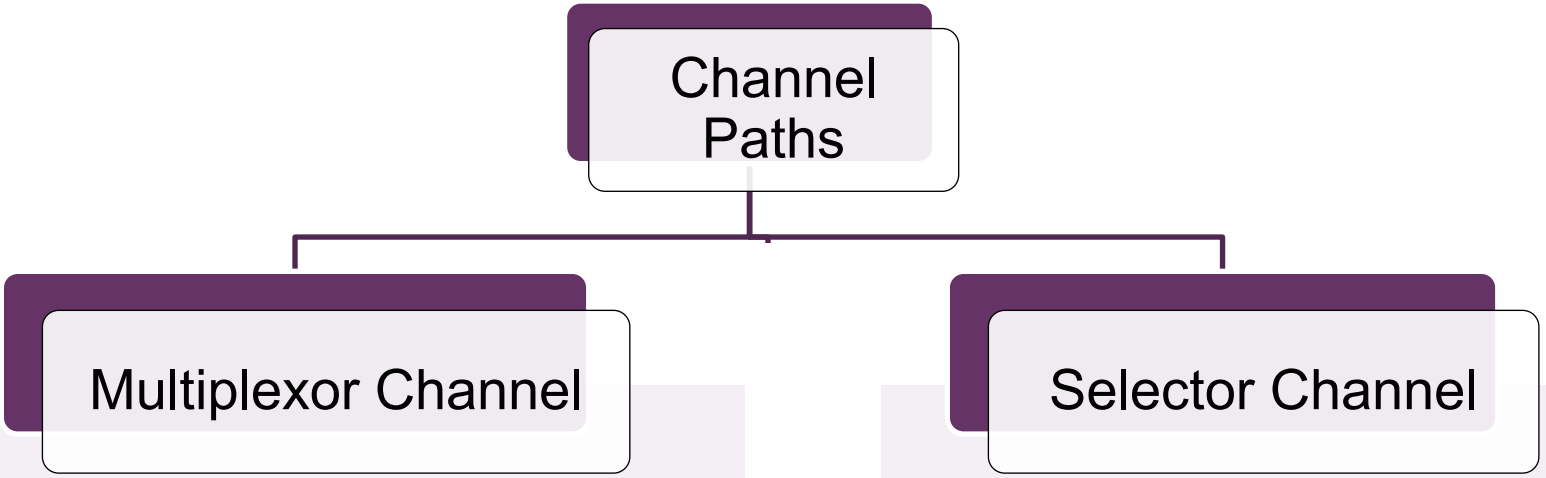


→ fine for small, single-user systems;



→ not scale well to large, multi-user systems, *e.g.* mainframe computers.

- Channel I/O → consists of one or more *I/O processors* (IOPs) that control various I/O pathways (*channel paths*).



Channel paths for "slow" devices can be combined (*multiplexed*) under one controller with faster channel.

(e.g. terminals, printers)

Channel paths for "fast" devices.

(e.g. disk drives)

1 – IOPs = small CPUs.

2 – I/O buses – help to isolate the host from the I/O operation.

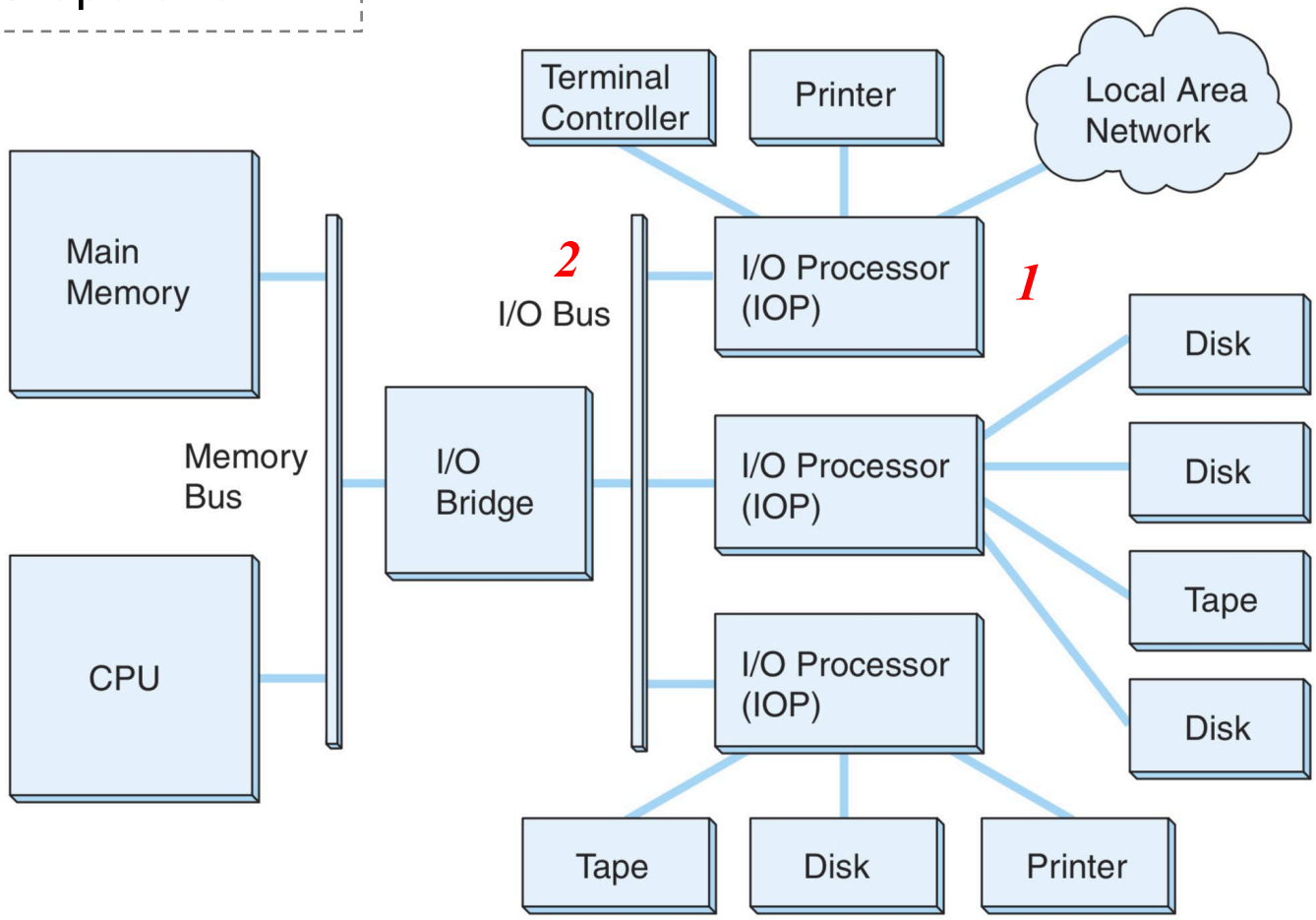


Figure: A channel I/O configuration.

1 – IOPs = small CPUs.

2 – I/O buses – help to isolate the host from the I/O operation.

Example:

Copying files from **disk** to **tape**, IOP uses only system memory bus to fetch its instruction from main memory.

The remainder transfer is affected using only I/O bus that controlled by IOP.

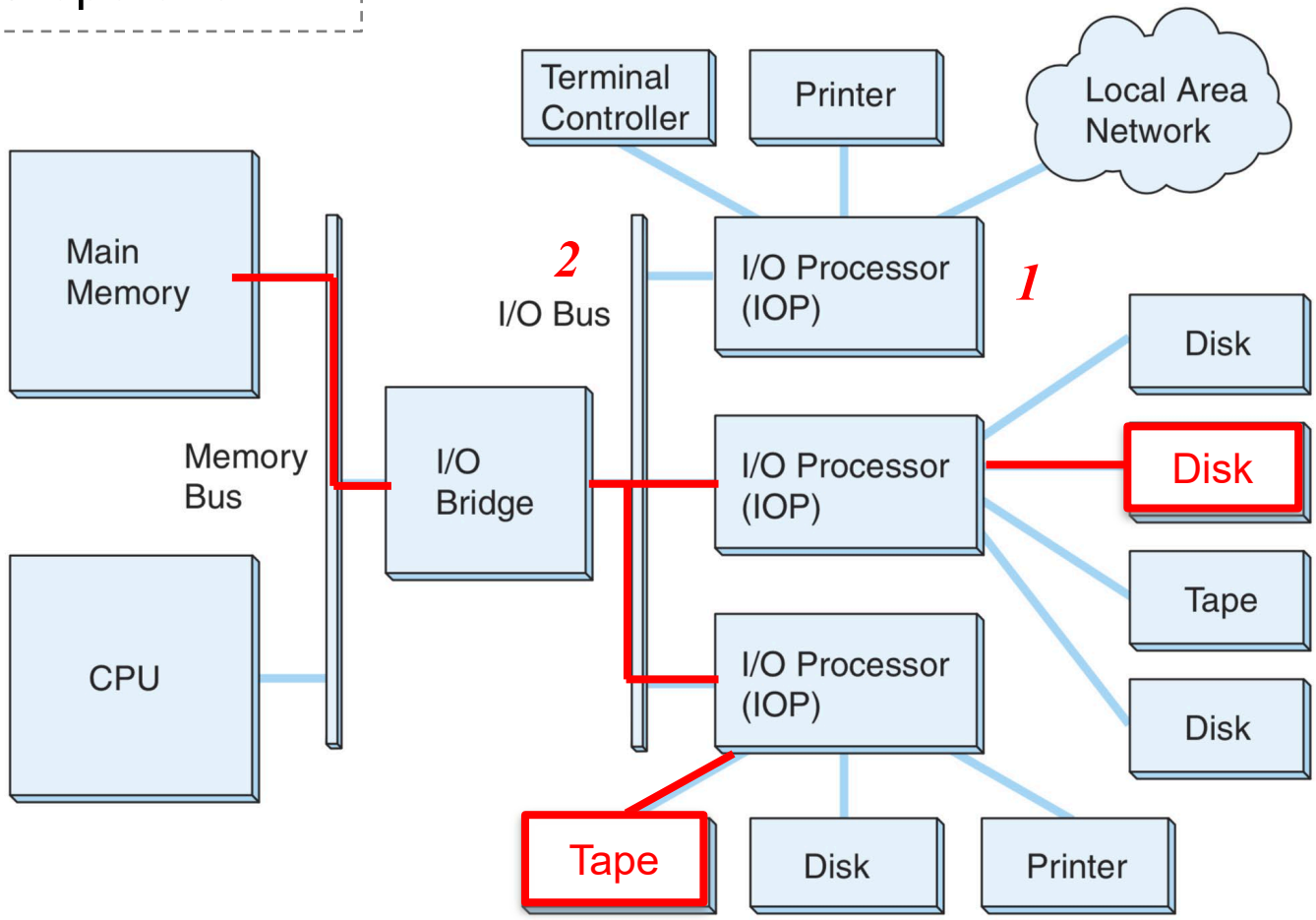
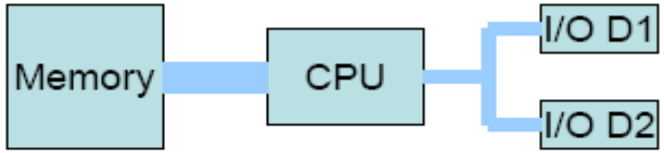


Figure: A channel I/O configuration.

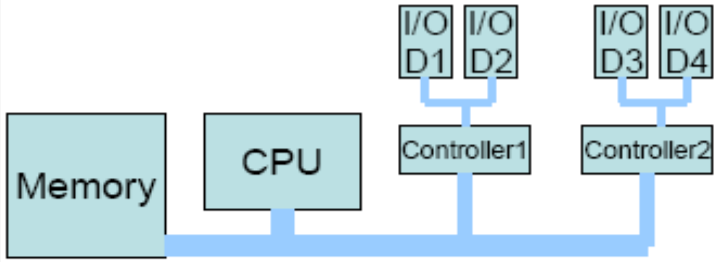
I/O Control Methods: A Comparison

Programmed & Interrupt Driven I/O



1. CPU controlled
2. All I/O goes through the CPU
3. No direct communication between the memory and the I/O devices
4. No intermediaries between the CPU and the I/O devices. CPU directly controls the I/O devices
5. I/O done in bytes i.e., each CPU instruction sends/receives one byte at a time

DMA & Channel I/O



1. CPU delegates control to devices controllers
2. Direct communication between I/O controllers and the memory
3. I/O done in blocks/files

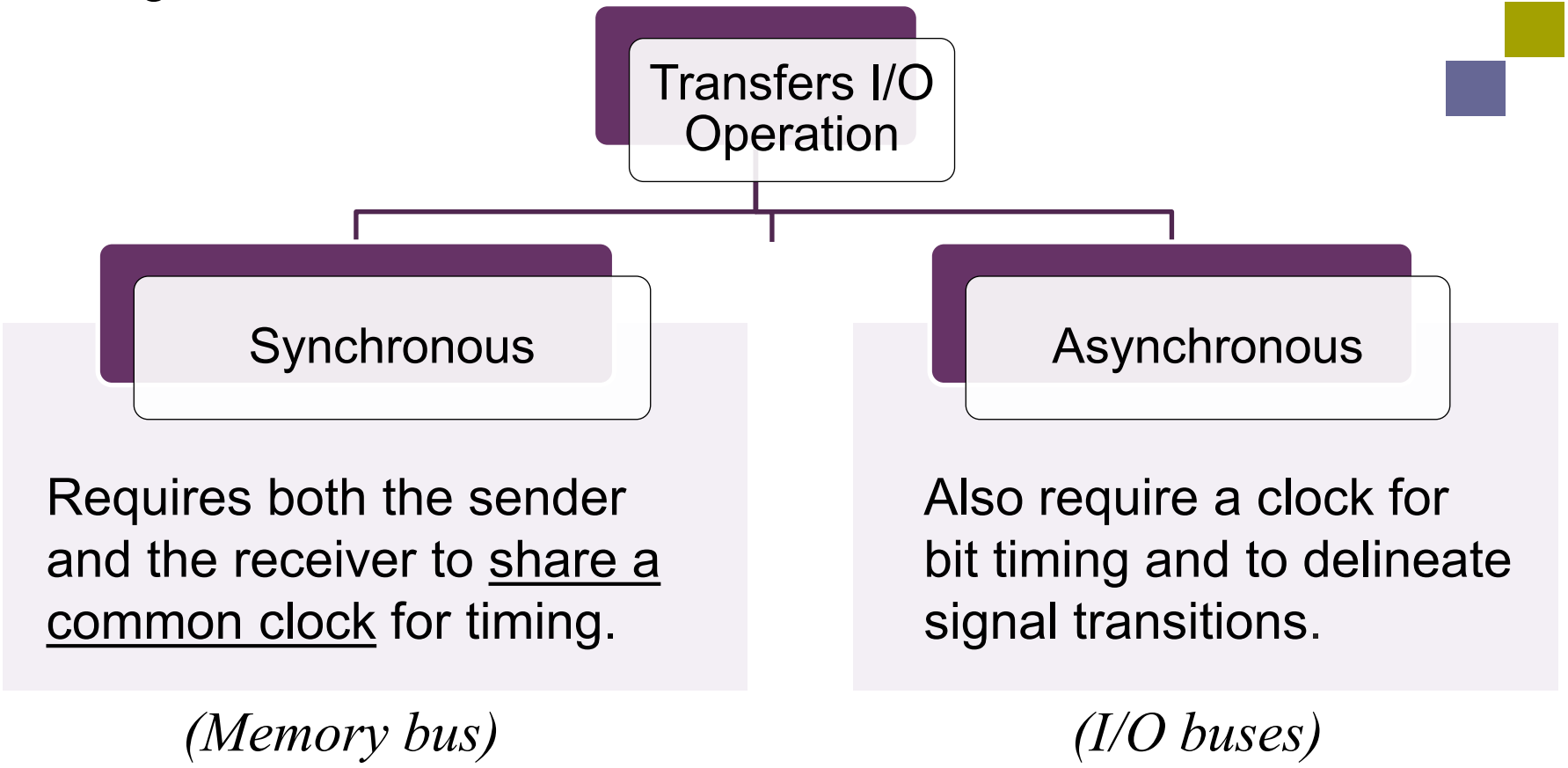
I/O Bus Operation

- The memory bus and the I/O bus can be separate entities.
- One good reason for having memory on its own bus:
  - memory transfers can be *synchronous*, using some multiple of the CPU's clock cycles to retrieve data from main memory.



I/O buses, on the other hand, cannot operate synchronously because I/O devices cannot always be ready to process an I/O transfer.

- In general:



To distinguish *synchronous* from *asynchronous* transfers will become clear after we look at the next example.

- The connection between the DMA circuit and the device interface circuits is more accurately represented by the following figure, which shows the individual component buses.

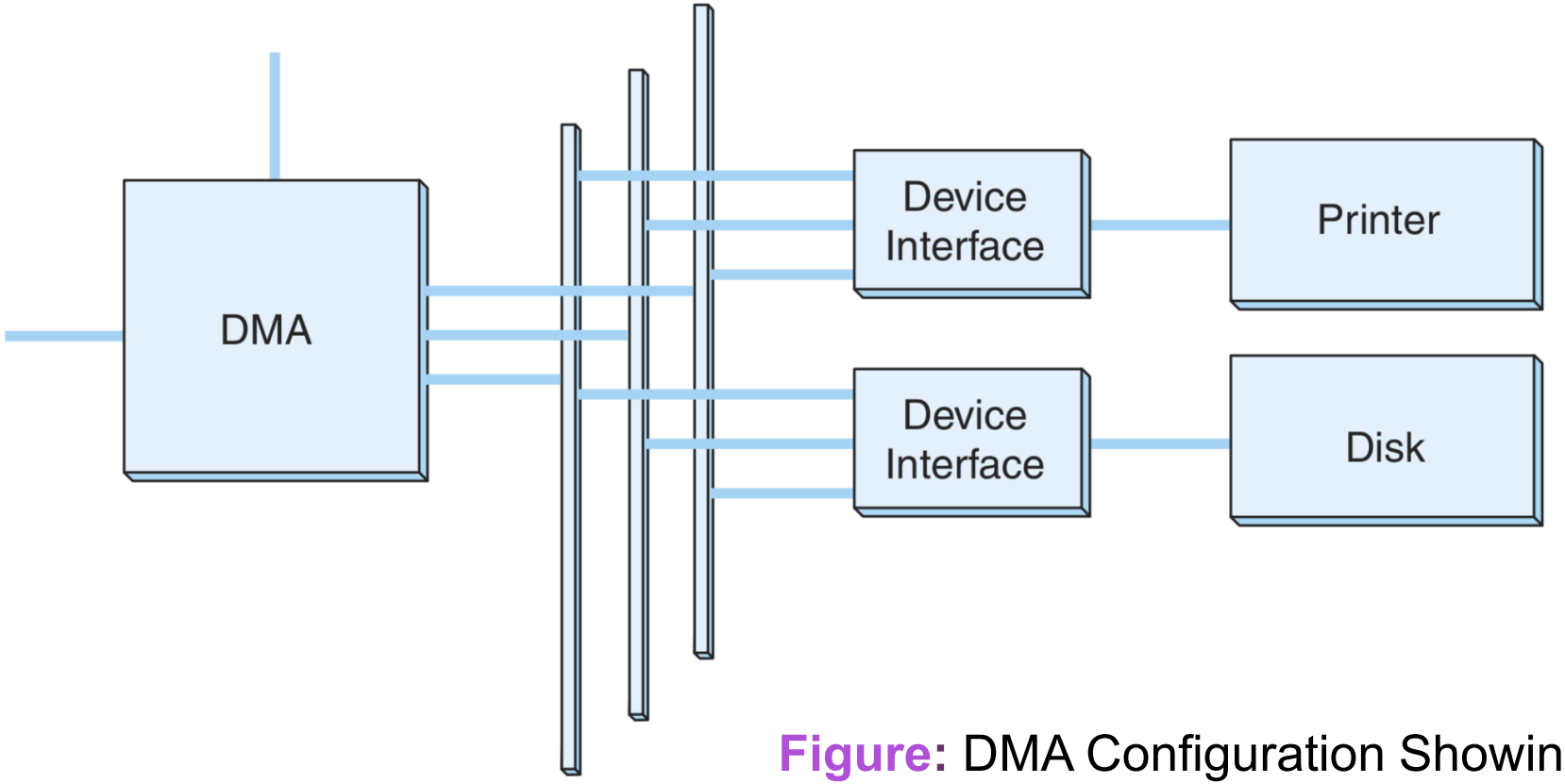


Figure: DMA Configuration Showing Separate Address, Data, and Control Buses.

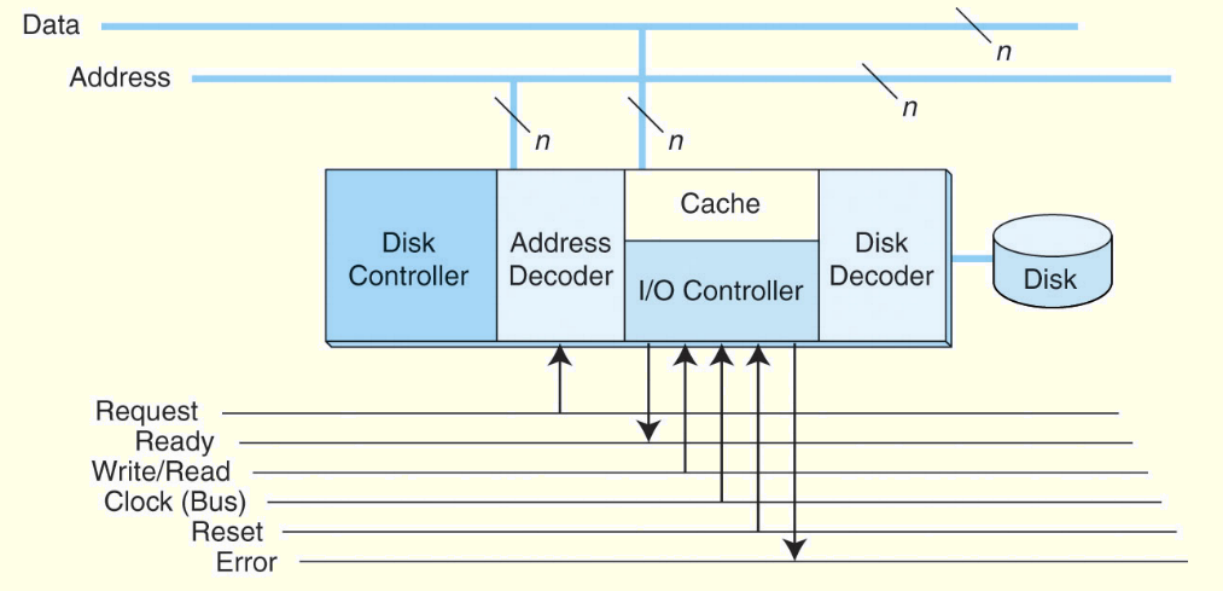
DMA bus connection to I/O device

This is how a DMA bus connects to a disk drive:

Data buses - width of the bus (i.e., 8 lines = 1 byte data at a time)

Address bus – can uniquely identify each device on the bus.

Control buses - coordinate activities of the bus and its attached devices.



Module 7
I/O and Storage Systems

7.1 Introduction
7.2 I/O Architectures
7.3 Data Storage
7.4 Summary

- Overview
- Magnetic Disk Technology
- Optical Disks
- Magnetic Tape
- RAID

7

Overview

- This section examines a range of external memory devices and systems.
- It begins with the most important device, the magnetic disk.
- The external optical memory is also examined, then magnetic tape is described.
- Finally, examines the use of disk arrays to achieve greater performance, looking specifically at the family of systems known as RAID (*Redundant Array of Independent Disks*).

(1) Magnetic Disk Technology

- *Magnetic disks as the most important device are the foundation of external memory on virtually all computer systems.
- Magnetic disks offer large amounts of durable storage that can be accessed quickly.
- Disk drives are called *random* (sometimes *direct*) access devices because each unit of storage, the *sector*, has a unique address that can be accessed independently of the sectors around it.
- Magnetic disk organization is shown on the following slide.

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.287.
*William Stallings (2016). *Computer Organization and Architecture: Designing for Performance* (10th Edition). United States: Pearson Education Limited, p.195.

- *Sectors* are divisions of concentric circles called *tracks*.

Disk tracks are consecutively numbered starting with track 0 at the outermost edge of the disk.

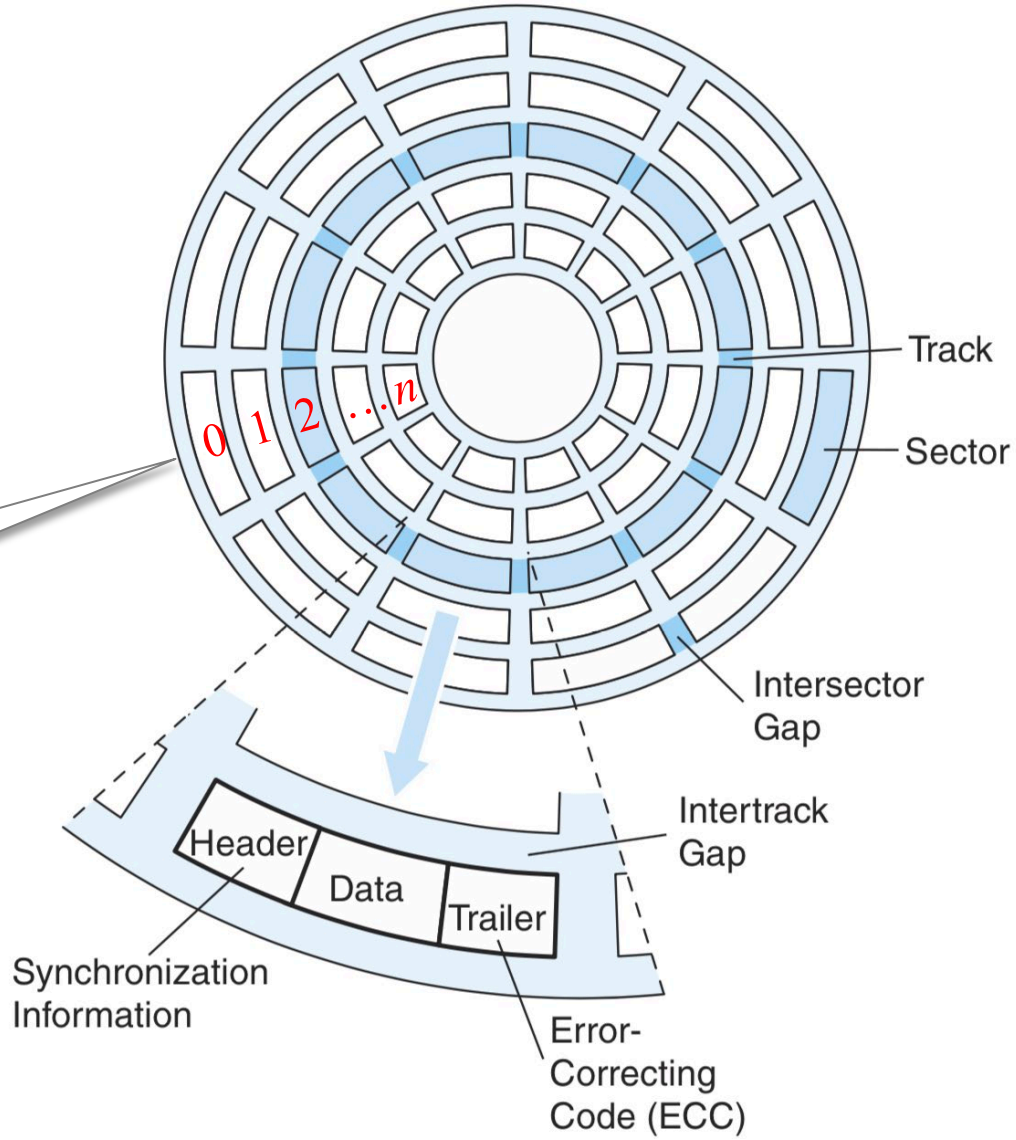
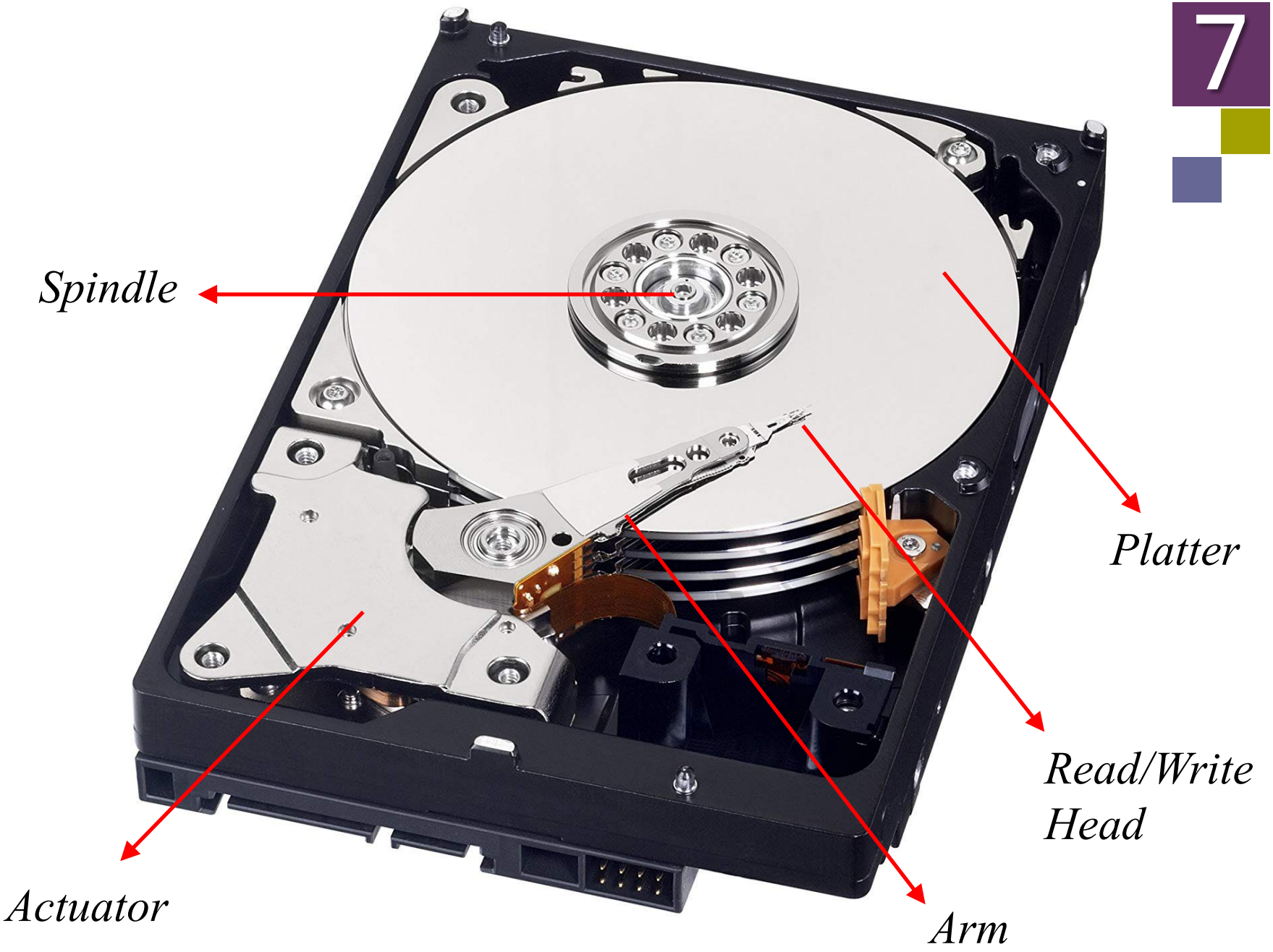


Figure: Disk Sectors Showing Intersector Gaps and Logical Sector Format.



https://images-na.ssl-images-amazon.com/images/I/81SYu2aMXRL._SL1500_.jpg (5 Jan 2019)

Flexible (Floppy) Disks

- Flexible disks are organized in much the same way as hard disks, with addressable *tracks* and *sectors*.
- Physical and logical limitations restrict floppies to much lower densities than hard disks.
- A major logical limitation of the DOS/Windows floppy diskette is the organization of its *File Allocation Table* (FAT).



Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.292.
https://www.silicon.co.uk/wp-content/uploads/2014/05/shutterstock_166068341.jpg (5 Jan 2019)

(2) Optical Disks



- Optical storage systems offer (practically) unlimited data storage at a cost that is competitive with tape.
- Optical disks come in a number of formats, the most popular format being the ubiquitous CD-ROM (*Compact Disk-Read Only Memory*).

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.293.
http://www.ommdvd.com/images/services/Duplication-Optical-Media-sm.jpg (5 Jan 2019)

- Variation of optical disks: CD-R (*CD-Recordable*), CD-RW (*CD-Rewritable*), and WORM (*Write Once Read Many*) disks are optical storage devices often used for :
  - long-term data archiving and
  - high-volume data output.

- For long-term archival storage of data, some computer systems send output directly to optical storage rather than paper or microfiche → COLD (*Computer Output Laser Disk*)



CD-ROM

(*Compact Disk-Read Only Memory*)



The total capacity data is 650MB

- CD-ROMs are polycarbonate (plastic) disks which a reflective aluminium film is applied.
- Compact disks are written from the center to the outside edge using a single spiraling track of bumps in the polycarbonate substrate.
- CD-ROMs were designed for storing music and other sequential audio signals.

DVD

(*Digital Versatile Disk*)



**The total capacity data are 17GB (double sided, dual-layer).*

- DVDs (formerly called *digital video disks*), can be thought of as quad-density CDs.
- DVDs rotate at about three times faster than CDs.
- Unlike CDs, DVDs can be single-sided or double-sided, called *single layer* or *double layer*.
- One can expect that DVDs will eventually replace CDs for long-term data storage and distribution.

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.297-298.
*William Stallings (2016). *Computer Organization and Architecture: Designing for Performance* (10th Edition). United States: Pearson Education Limited, p.221.

High-Definition (HD) Optical Disks





- Designed to store high-definition videos and to provide significantly greater storage capacity compared to DVDs
- Two competing disk formats and technologies :

Blue-Ray DVD	HD-DVD
Can store 25GB on a single layer on a single side.	Can store 15GB on a single layer on a single side.

The big difference → HD-DVD is *backward compatible* with red laser DVDs, while Blu-Ray is not.

(3) Magnetic Tape



- Magnetic tape is the oldest and most cost-effective of all mass-storage devices.
- First-generation magnetic tapes were made of the same material used by analog tape recorders.

Early tapes had capacities under 11MB, and required nearly a half hour to read or write the entire reel.

7

(4) RAID

(*Redundant Array of Independent Disks*)

- RAID → invented to address problems of disk reliability, cost, and performance.
- The disk storage can gains in performance by developing an array of disks that operates independently using multiple parallel components.



How it work?

- data is stored across many disks.
- extra disks added to the array to provide error correction (redundancy).

Linda Null and Julia Lobur (2003). *The Essentials of Computer Organization and Architecture.* United States: Jones and Bartlett Publishers. p.301.
William Stallings (2016). *Computer Organization and Architecture: Designing for Performance* (10th Edition). United States: Pearson Education Limited, p.204.

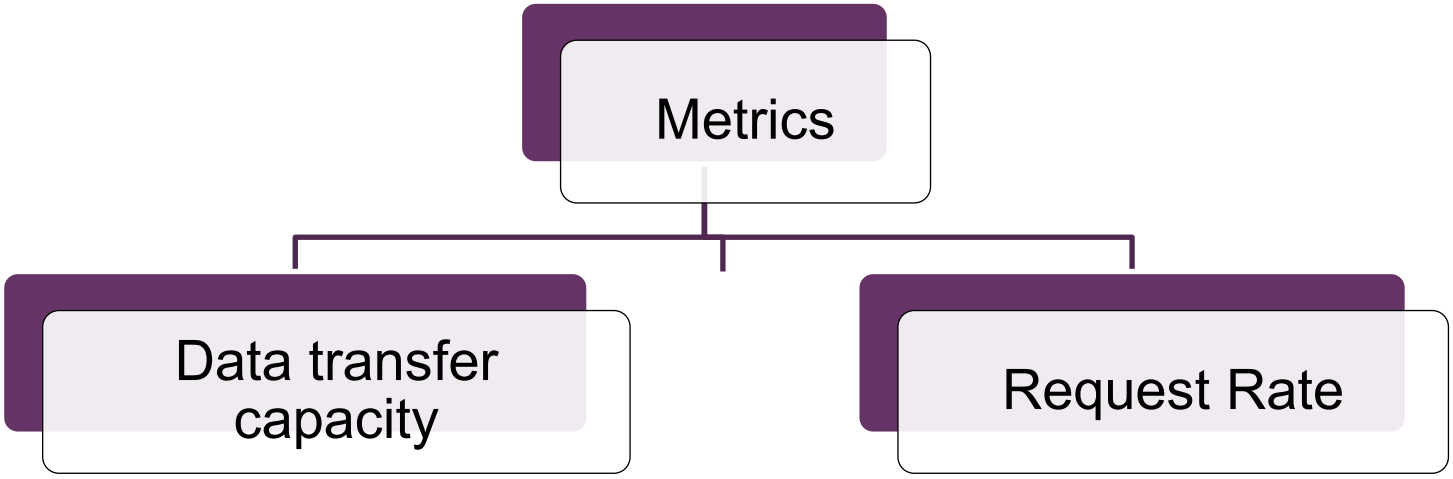
- There are a wide variety of ways in which data can be organized and which redundancy can be added to improve reliability in multiple disks.
- There are 7 types (called *levels 0 – 6*) of RAID, each having different performance and reliability of 3 common characteristics:

RAID → a set of physical disk drives viewed by the OS as a single logical drive.

Data → distributed across the physical drives of an array.

Redundant disk → used to store parity information to guarantee data recoverability when disk failure.

- The RAID levels inherently perform differently relative to TWO metrics of I/O performance:



- The next table provides a rough guide to the seven levels.

Table: RAID levels.

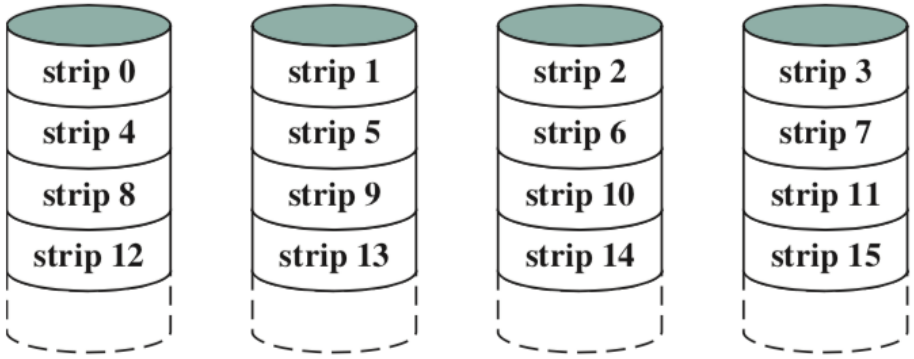
Category	Level	Description	Disks Required	Data Availability	Large I/O Data Transfer Capacity	Small I/O Request Rate
Striping	0	Nonredundant	N	Lower than single disk	Very high	Very high for both read and write
Mirroring	1	Mirrored	$2N$	Higher than RAID 2, 3, 4, or 5; lower than RAID 6	Higher than single disk for read; similar to single disk for write	Up to twice that of a single disk for read; similar to single disk for write
Parallel access	2	Redundant via Hamming code	$N + m$	Much higher than single disk; comparable to RAID 3, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
	3	Bit-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
Independent access	4	Block-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 5	Similar to RAID 0 for read; significantly lower than single disk for write	Similar to RAID 0 for read; significantly lower than single disk for write
	5	Block-interleaved distributed parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 4	Similar to RAID 0 for read; lower than single disk for write	Similar to RAID 0 for read; generally lower than single disk for write
	6	Block-interleaved dual distributed parity	$N + 2$	Highest of all listed alternatives	Similar to RAID 0 for read; lower than RAID 5 for write	Similar to RAID 0 for read; significantly lower than RAID 5 for write

Note: N = number of data disks; m proportional to log N

Darker shading indicate the strong point for each level

RAID Level 0

(*Striping*)



(a) RAID 0 (Nonredundant)

- Not a true of the RAID family because it has no redundancy to improve reliabilty.
- Some applications are concerns on performance and capacity, but the low cost is more important than improved reliability.

- In RAID 0, the user and system data are distributed across all the disks in the array.

- All users and system data are viewed as being stored on ONE logical disk.
- The logical disk is divided into *strips* (*e.g.* physical blocks, sectors, or some other unit).
- The strips are mapped round-robin to consecutive physical disks in the RAID array (Refer next figure for data mapping).



Simple data distribution across the disk array.



Low reliability when failure of any disk.

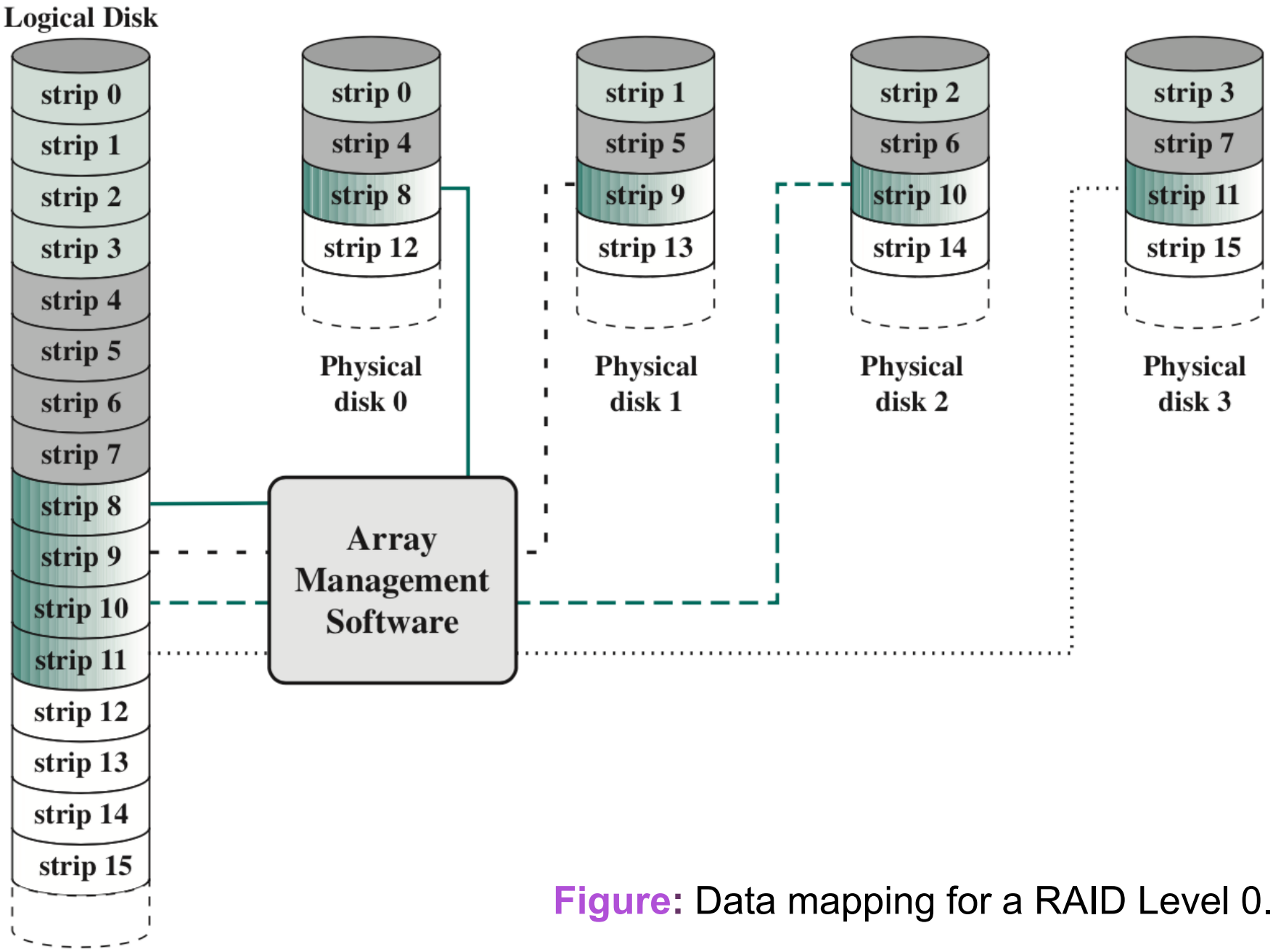
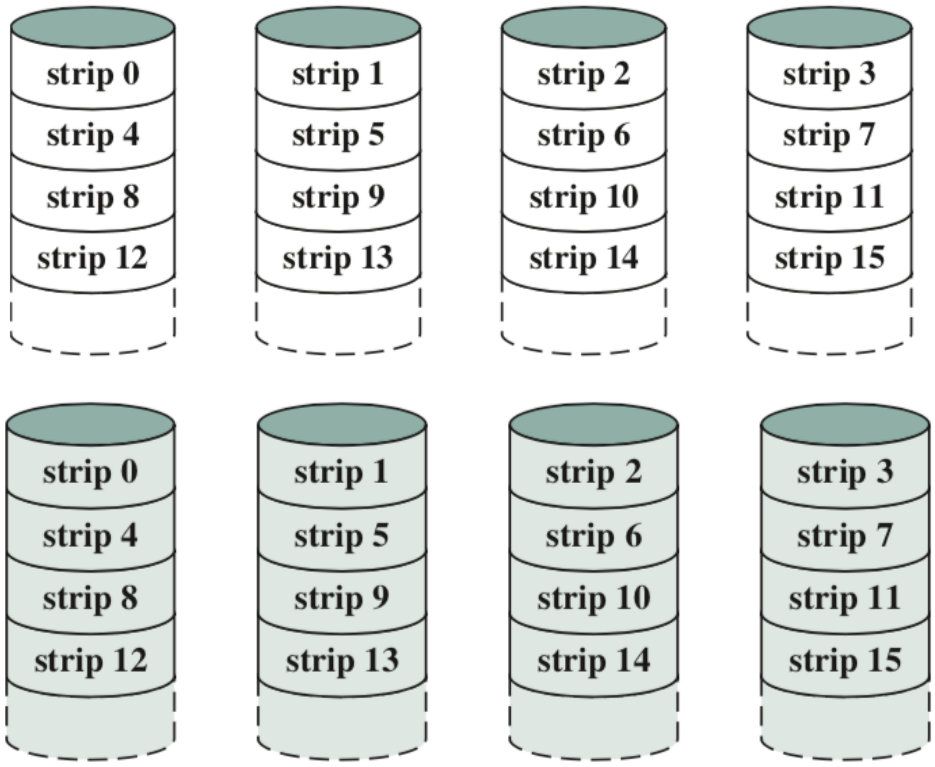


Figure: Data mapping for a RAID Level 0.

RAID Level 1

(*Mirrored*)

- Some form of parity calculation is used to introduce redundancy (duplication of all data).
- Each logical strip is mapped to two separate physical disks.



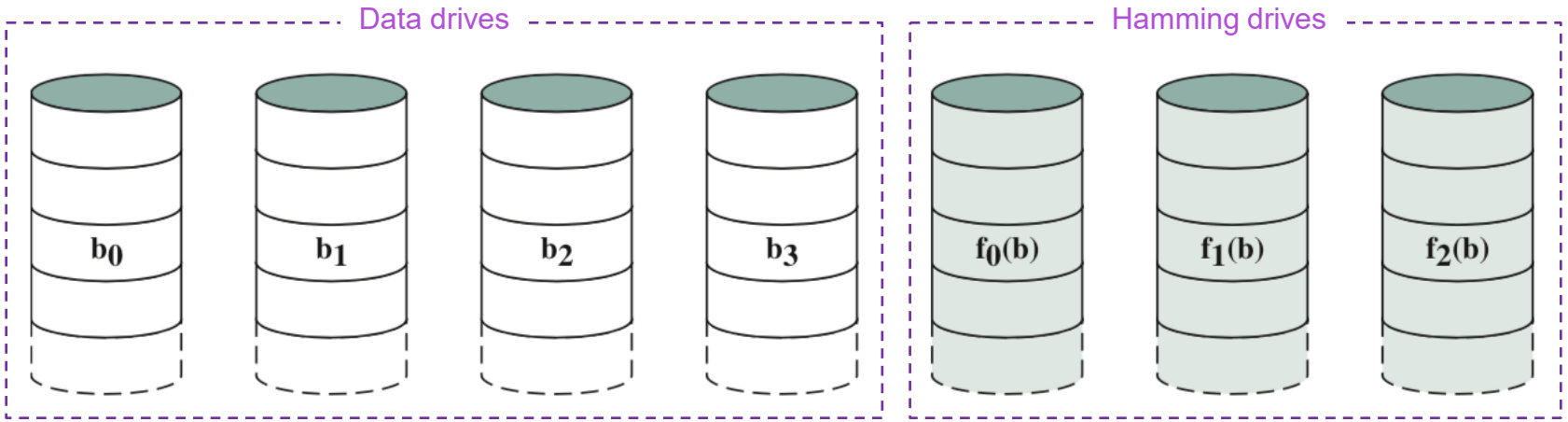
(b) RAID 1 (Mirrored)

100% redundancy and good performance.

Cost is expensive.

RAID Level 2

(*Parallel access*)



(c) RAID 2 (Redundancy through Hamming code)

- This RAID 2 make uses of a parallel access technique.
- All members disks participate in the execution of every I/O request.

- Typically, Hamming code is used for error correction.

  - An error-correcting code is calculated across corresponding bits on each data disk (*Data drives*);
  - The bits of the code are stored in the corresponding bit positions of multiple parity disks (*Hamming drives*).



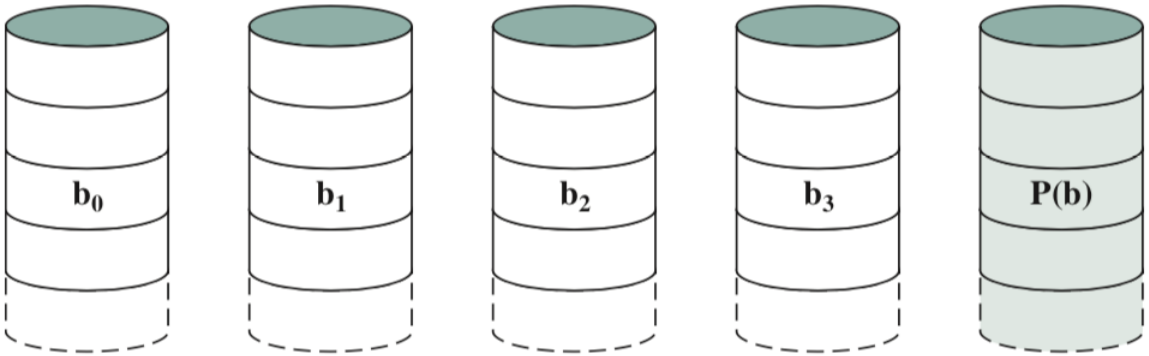
Provides error correction.



Performance is poor and the cost is relatively high.

RAID Level 3

(*Parallel access*)



(d) RAID 3 (Bit-interleaved parity)

- Organized similarly to RAID 2.
- RAID 3 requires only a single redundant parity disk.
- Parity is the XOR of the data bits.

Instead of an error-correcting code, a simple *parity bit* is computed for the set of individual bits in the same position on all data disks.



- **Redundancy**: In the event of a data drive failure, the parity drive is accessed and data is reconstructed from the remaining devices.
- **Performance**: Can achieve very high data transfer rates due to the data striped on very small strips.



- Consistency of the parity must be maintained for later regeneration.
- Not suitable for commercial applications, but good for personal systems.

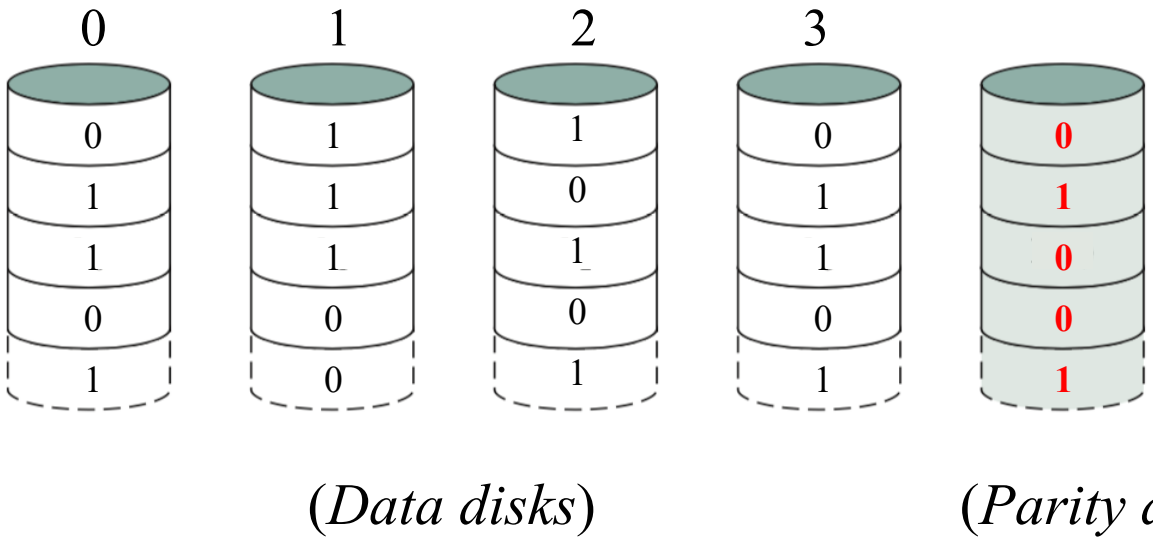


A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

Example :

To write (a) the parity bits and (b) recover the data drive 2.

(a)



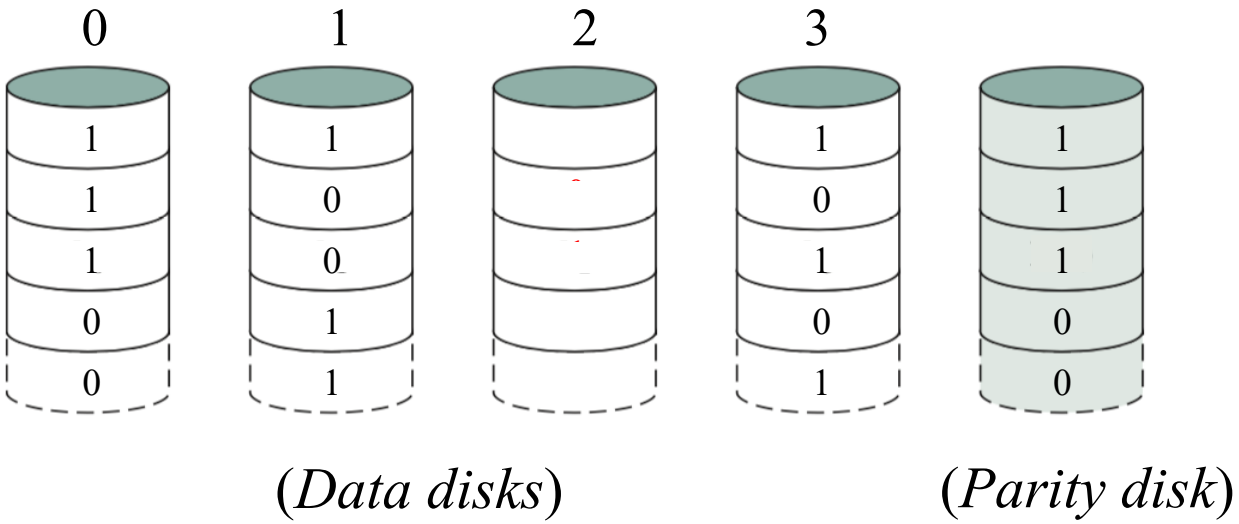


A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

Example :

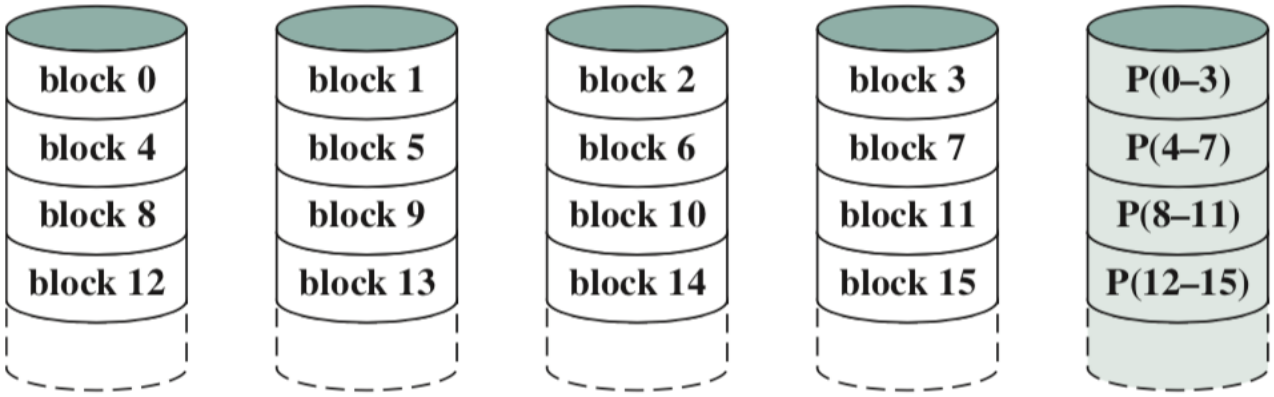
To write (a) the parity bits and (b) recover the data drive 2.

(b)



RAID Level 4

(*Independent access*)



(e) RAID 4 (Block-level parity)

- This RAID 4 make uses of an independent access technique.
- Each member disk operates independently so that separate I/O requests can be satisfied in parallel.

- Strips are relatively large in RAID 4.

  - A bit-by-bit parity strip is calculated across corresponding strips on each data disk (*Data drives*);
  - The parity bits are stored in the corresponding strip on the parity disk (*Parity drives*).



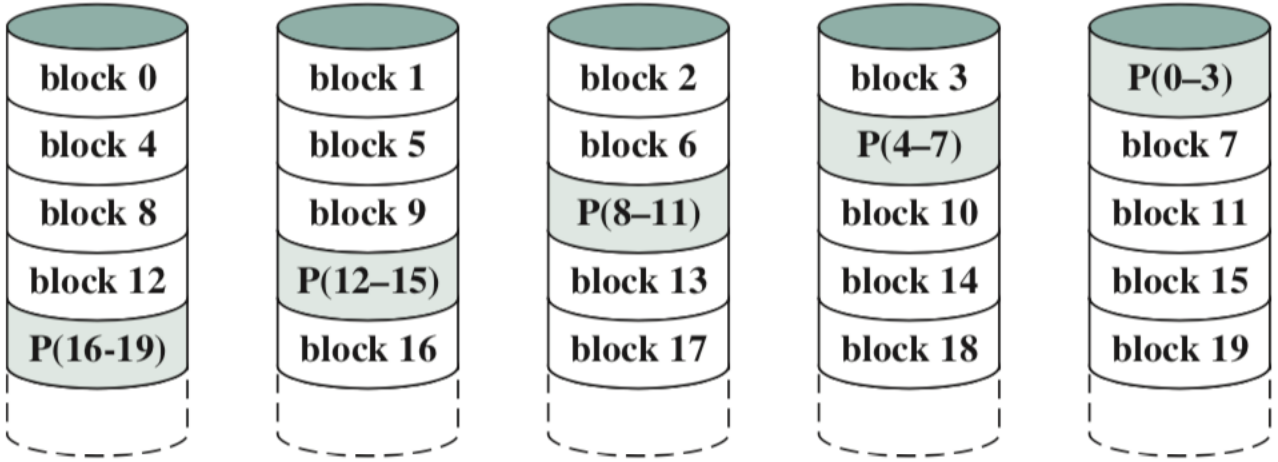
Suitable for applications that require high I/O request rates.



Less suited for applications that require high data transfer rates because each write operation involve the parity disk (bottleneck).

RAID Level 5

(*Independent access*)

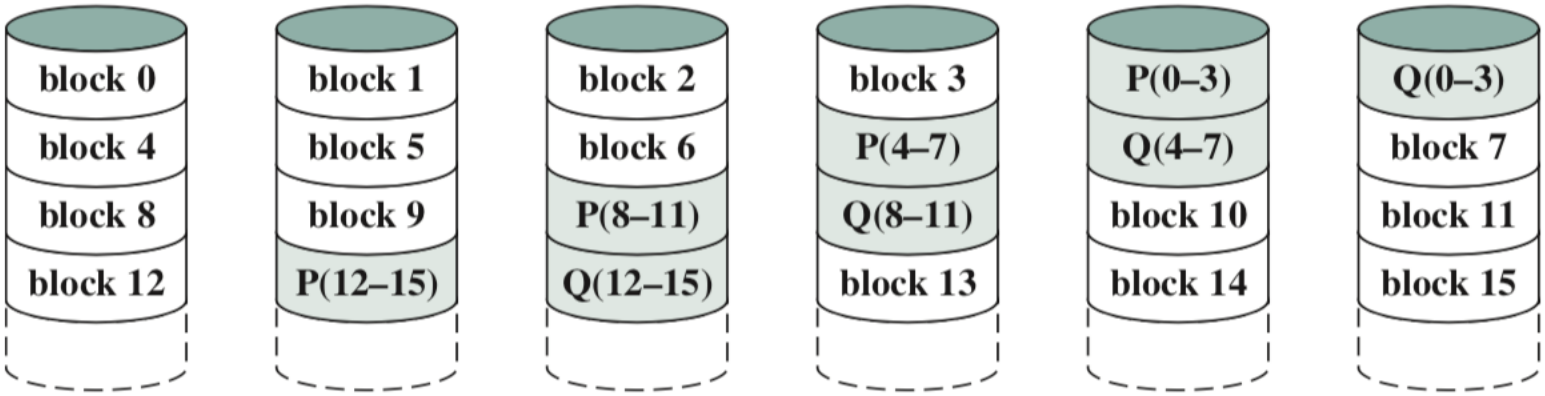


(f) RAID 5 (Block-level distributed parity)

- Similar to RAID 4 but this distributes the parity strips across all disks with a typical allocation (round-robin scheme).
- The distribution of parity strips avoids the potential bottleneck found in RAID 4.

RAID Level 6

(*Independent access*)



(g) RAID 6 (Dual redundancy)

- Two different parity calculations are carried out and stored in separate blocks on different disks.
- This make it possible to regenerate data even if two disks containing user data fail.



Provides extremely high data availability.



Suffer more than 30% drop in overall write performance compared with a RAID 5.

** RAID 5 and RAID 6 read performance is comparable.*

7.4 Summary

7

- I/O systems are critical to the overall performance of a computer system.
  - consist of memory blocks, cabling, control circuitry, interfaces, and media.
- I/O control methods include programmed I/O, interrupt-based I/O, DMA, and channel I/O.
- Buses require control lines, a clock, and data lines. Timing diagrams specify operational details.



- Magnetic disk is the principal form of durable storage.
- Optical disks provide long-term storage for large amounts of data, although access is slow.
- Magnetic tape is also an archival medium. Recording methods are track-based, serpentine, and helical scan.

- RAID gives disk systems improved performance and reliability.